The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Ambient Shell Law
Necessity / Uniqueness from Combined Response Substrate Law

Abstract

The current deepest benchmark-facing note on the active primitive-reservoir observer-reduction line already changed the burden once: the primitive reversible constrained substrate dynamics normal form is no longer merely available, because it is derived canonically from an ambient shell-law package with hidden-sector gap and is necessary there up to the forced translated-shell and orthogonal-hidden identifications. The present note goes one honest layer deeper again. The remaining task is no longer to justify that primitive normal form, but to justify why the ambient shell-law class itself is forced on the active line rather than becoming the new stopping point.

The new theorem does that by compressing the ambient shell-law package into a single deeper combined response substrate law. For each sector it starts from a compact convex primitive substrate body, one affine combined response map to observer data, shell response, and hidden response, a product-factorized response image, an exact primitive-shell realization on the zero-response locus, symmetry and time-reversal actions preserving the combined response image, a positively gapped hidden factor, shell solvability on fixed data fibers, and a response-level coercivity condition on data-invisible variations. From that single package one derives canonically the ambient state space and ambient body, the affine observer-data readout, the ambient shell plane, the affine shell law and shell value, the hidden-sector splitting and positive gap operator, the symmetry and time-reversal structure, the shell solvability property, and the shell-direction coercivity statement.

The theorem is stronger than another deeper existence relay. On this combined-response line the ambient shell-law package is necessary: the ambient body is the combined-response image body, the readout and shell law are the coordinate projections, the shell plane is the hidden-zero plane, the shell value is the zero response value, and the hidden sector is the hidden response factor with its given gap operator. After a canonical barycentric normalization of the shell-plane reference point, no further ambient freedom remains except the obvious factor-preserving affine identification. The downstream benchmark package stays fixed throughout: the unique selector defined by the zero-response shell locus still pulls back exactly the same one operative metric $g^\sharp$ and exactly the same ordinary-matter route through that metric, with no second operative metric and no enlarged low-energy matter law. This remains a disciplined derivational gain rather than a completed final microphysical derivation of GR.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has again narrowed the remaining honest burden. The observer-reduction datum line, the defect-fiber factorization theorem, the one-metric output theorem, the chain-forcing theorem, the selector-existence theorem,

the stationary-construction theorem, the explicit-package necessity / uniqueness theorem, and the later ambient-shell-law theorem have each discharged what was honestly open at their respective layers. The most recent of those notes matters because it alters what now counts as primary progress. Once the primitive reversible constrained dynamics normal form is itself forced by a deeper ambient shell-law package, another note that merely restates that ambient package would no longer count as the next front-line move.

The present note therefore addresses the burden that remained open after the ambient-shell-law theorem itself. The task is now to explain why that ambient shell-law class should hold on the active line. The positive move made here is to collapse that class into a still more primitive combined response substrate law in which the observer-data channel, shell response, and hidden response appear together as one affine response image of a compact convex primitive substrate body.

This note does not alter the benchmark package. It does not change the one operative completed metric, it does not change the ordinary matter route, it does not introduce a second operative metric, and it does not enlarge the low-energy matter law. Its gain is upstream and structural: the ambient shell-law package is shown to be a canonical and necessary output of a deeper combined response substrate law with hidden-sector gap.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim a completed ultimate microphysical theory. Its narrower theorem is only that, on the active primitive-observer-reduction line, the ambient shell-law package is itself a forced and canonically normalized output of a deeper combined response substrate law.

2. Fixed Primitive Continuation Data

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}^{\mathrm{for}}_{\mathrm{car}} \times \mathcal{I}^{\mathrm{for}}_{\mathrm{cur}} \times \mathcal{I}^{\mathrm{for}}_{\mathrm{hom}}.$$

3. *Fixed shell extractors*

$$\mathscr{E}^{\mathrm{cur}}_{\sigma} : Z^{\mathrm{cur}}_{\sigma} \to Y^{\mathrm{cur}}_{\sigma}, \qquad \mathscr{E}^{\mathrm{eq}}_{\sigma} : Z^{\mathrm{eq}}_{\sigma} \to Y^{\mathrm{eq}}_{\sigma},$$

together with the recorded shell composites

$$\mathcal{J}^{\mathrm{prim}}_{\sigma} = \mathscr{E}^{\mathrm{cur}}_{\sigma} \circ \mathfrak{C}^{\mathrm{cur}}_{\sigma}, \qquad \Theta^{\mathrm{prim}}_{\sigma} = \mathscr{E}^{\mathrm{eq}}_{\sigma} \circ \mathfrak{C}^{\mathrm{eq}}_{\sigma}. \tag{1}$$

4. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^{\sharp}, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T^{\mathrm{matter}}_{\mu\nu}[g^{\sharp}, \psi] \tag{2}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Remark 1. *Definition 1 is not reopened here. The primitive shell data and the benchmark one-metric / ordinary-matter outputs remain fixed continuation data. The new burden begins one layer deeper again: why the ambient shell-law package that previously forced the primitive normal form should itself be forced.*

3. Target Ambient Shell-Law Package

Definition 2 (Ambient shell law with hidden-sector gap)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *An* ambient shell-law package with hidden-sector gap *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean ambient state space*

$$\mathcal{A}_{\sigma}$$

with a compact convex body

$$P_{\sigma} \subset \mathcal{A}_{\sigma}.$$

2. *A smooth embedding*

$$\jmath_{\sigma} : X_{\sigma} \hookrightarrow \widehat{X}_{\sigma}.$$

3. *An affine observer-data readout*

$$\Omega_{\sigma} : \mathcal{A}_{\sigma} \to \widehat{X}_{\sigma} \times Z^{\mathrm{cur}}_{\sigma} \times Z^{\mathrm{eq}}_{\sigma}.$$

4. *An affine shell plane*

$$\mathcal{A}^{\mathrm{sh}}_{\sigma} \subset \mathcal{A}_{\sigma}$$

together with a distinguished reference point

$$a^{0}_{\sigma} \in \mathcal{A}^{\mathrm{sh}}_{\sigma} \cap P_{\sigma}.$$

Assume moreover that

$$a_\sigma^0 \in \operatorname{int}_{\mathcal{A}_\sigma^{\mathrm{sh}}}(P_\sigma \cap \mathcal{A}_\sigma^{\mathrm{sh}}).$$

Write

$$V_\sigma^{\mathrm{sh}} \equiv T\mathcal{A}_\sigma^{\mathrm{sh}}, \qquad N_\sigma^{\mathrm{hid}} \equiv (V_\sigma^{\mathrm{sh}})^\perp,$$

and let

$$\Pi_\sigma^{\mathrm{sh}} : \mathcal{A}_\sigma \to V_\sigma^{\mathrm{sh}}, \qquad \Pi_\sigma^{\mathrm{hid}} : \mathcal{A}_\sigma \to N_\sigma^{\mathrm{hid}}$$

denote the orthogonal projections.

5. *An affine shell law*

$$\Gamma_\sigma : \mathcal{A}_\sigma \to E_\sigma^{\mathrm{amb}}$$

and a distinguished shell value

$$\gamma_\sigma \in E_\sigma^{\mathrm{amb}}.$$

6. *A compact symmetry group*

$$\mathbb{G}_\sigma \subset O(\mathcal{A}_\sigma)$$

and an orthogonal involution

$$\vartheta_\sigma \in O(\mathcal{A}_\sigma)$$

preserving P_σ, $\mathcal{A}_\sigma^{\mathrm{sh}}$, a_σ^0, Ω_σ, Γ_σ, and γ_σ.

7. *Body splitting over shell and hidden directions: the map*

$$a \longmapsto (a_\sigma^0 + \Pi_\sigma^{\mathrm{sh}}(a - a_\sigma^0),\, \Pi_\sigma^{\mathrm{hid}}(a - a_\sigma^0)) \tag{3}$$

is a bijection from P_σ onto

$$(P_\sigma \cap \mathcal{A}_\sigma^{\mathrm{sh}}) \times \Pi_\sigma^{\mathrm{hid}}(P_\sigma - a_\sigma^0).$$

Assume also that

$$0 \in \operatorname{int} \Pi_\sigma^{\mathrm{hid}}(P_\sigma - a_\sigma^0).$$

8. *Hidden invisibility of the observable shell law and data readout: for every*

$$s \in P_\sigma \cap \mathcal{A}_\sigma^{\mathrm{sh}}, \qquad n \in \Pi_\sigma^{\mathrm{hid}}(P_\sigma - a_\sigma^0),$$

whenever $s + n \in P_\sigma$,

$$\Omega_\sigma(s+n) = \Omega_\sigma(s), \qquad \Gamma_\sigma(s+n) = \Gamma_\sigma(s). \tag{4}$$

9. *Exact primitive-shell realization: the shell locus*

$$P_\sigma \cap \mathcal{A}_\sigma^{\mathrm{sh}} \cap \Gamma_\sigma^{-1}(\gamma_\sigma)$$

is the image of a smooth embedding

$$\eta_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{A}_\sigma$$

satisfying

$$\Omega_\sigma(\eta_\sigma(w)) = \big(\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)\big) \tag{5}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$.

10. *Shell solvability on fixed data fibers:*

$$\Omega_\sigma(P_\sigma) = \Omega_\sigma\big(P_\sigma \cap \mathcal{A}_\sigma^{\mathrm{sh}} \cap \Gamma_\sigma^{-1}(\gamma_\sigma)\big). \tag{6}$$

11. *Shell-direction coercivity: there exists* $\kappa_\sigma^{\mathrm{sh}} > 0$ *such that for every*

$$\delta q \in V_\sigma^{\mathrm{sh}} \quad \textit{with} \quad D\Omega_\sigma(a_\sigma^0)\,\delta q = 0,$$

one has

$$\big\|D\Gamma_\sigma(a_\sigma^0)\,\delta q\big\|^2 \geq \kappa_\sigma^{\mathrm{sh}}\,\|\delta q\|^2. \tag{7}$$

12. *A positive self-adjoint hidden-sector operator*

$$K_\sigma^{\mathrm{hid}} : N_\sigma^{\mathrm{hid}} \to N_\sigma^{\mathrm{hid}}$$

with lower bound

$$\Big\langle \eta, K_\sigma^{\mathrm{hid}}\eta \Big\rangle \geq \kappa_\sigma^{\mathrm{hid}}\,\|\eta\|^2 \qquad \textit{for all } \eta \in N_\sigma^{\mathrm{hid}} \tag{8}$$

for some $\kappa_\sigma^{\mathrm{hid}} > 0$*, and such that* $\mathbb{G}_\sigma$ *and* ϑ_σ *preserve* K_σ^{hid}.

Remark 2. *Definition 2 is the target package whose origin remained open after the previous note. The purpose here is not to reproduce it as another deeper assumption, but to show that it is a canonical output of still more primitive combined response data.*

4. Combined Response Substrate Law

Definition 3 (Combined response substrate law with hidden-sector gap)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* combined response substrate law with hidden-sector gap *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean primitive substrate space*

$$\mathcal{M}_\sigma$$

with a compact convex body

$$\mathcal{P}_\sigma \subset \mathcal{M}_\sigma.$$

2. *An observer-data target space*

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \equiv \widehat{X}_\sigma \times Z_\sigma^{\mathrm{cur}} \times Z_\sigma^{\mathrm{eq}},$$

an ambient shell target

$$E_\sigma^{\mathrm{amb}},$$

a hidden response space

$$H_\sigma,$$

and the product response space

$$\mathcal{R}_\sigma \equiv \mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus H_\sigma.$$

3. *An affine combined response map*

$$\Xi_\sigma = (\Xi_\sigma^{\mathrm{dat}}, \Xi_\sigma^{\mathrm{sh}}, \Xi_\sigma^{\mathrm{hid}}) : \mathcal{M}_\sigma \to \mathcal{R}_\sigma.$$

4. *The response image body*

$$\mathcal{B}_\sigma^{\mathrm{resp}} \equiv \Xi_\sigma(\mathcal{P}_\sigma) \tag{9}$$

is compact and convex, and its shell slice

$$\mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\})$$

has nonempty relative interior in

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\}.$$

5. *A compact convex hidden body*

$$D_\sigma \subset H_\sigma$$

with

$$0 \in \operatorname{int} D_\sigma,$$

such that the response image factorizes exactly as

$$\mathcal{B}_\sigma^{\mathrm{resp}} = \{(u, e, n) : (u, e, 0) \in \mathcal{B}_\sigma^{\mathrm{resp}}, \quad n \in D_\sigma\}. \tag{10}$$

Equivalently, the map

$$(u, e, n) \longmapsto ((u, e, 0), n) \tag{11}$$

is a bijection from $\mathcal{B}_\sigma^{\mathrm{resp}}$ *onto*

$$\Big(\mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\})\Big) \times D_\sigma.$$

6. *A positive self-adjoint hidden-sector operator*

$$K_\sigma^{\mathrm{hid}} : H_\sigma \to H_\sigma$$

with lower bound

$$\left\langle \eta, K_\sigma^{\mathrm{hid}} \eta \right\rangle \geq \kappa_\sigma^{\mathrm{hid}} \|\eta\|^2 \qquad \textit{for all } \eta \in H_\sigma \tag{12}$$

for some $\kappa_\sigma^{\mathrm{hid}} > 0$.

7. *A compact symmetry group*

$$\mathbb{H}_\sigma \subset O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}}) \times O(H_\sigma)$$

and an orthogonal involution

$$\rho_\sigma \in O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}}) \times O(H_\sigma)$$

preserving $\mathcal{B}_\sigma^{\mathrm{resp}}$*, the hidden-zero slice*

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\},$$

the zero shell value $0 \in E_\sigma^{\mathrm{amb}}$*, and* K_σ^{hid}*.*

8. *A smooth embedding*

$$\jmath_\sigma : X_\sigma \hookrightarrow \widehat{X}_\sigma$$

and a smooth shell realization map

$$\zeta_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{M}_\sigma$$

such that

$$\zeta_\sigma(\mathcal{I}_\sigma^{\mathrm{for}}) \subset \mathcal{P}_\sigma \cap (\Xi_\sigma^{\mathrm{sh}})^{-1}(0) \cap (\Xi_\sigma^{\mathrm{hid}})^{-1}(0),$$

the composition

$$\eta_\sigma \equiv \Xi_\sigma \circ \zeta_\sigma \tag{13}$$

is a diffeomorphism onto

$$\mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus \{0\} \oplus \{0\}),$$

and

$$\Xi_\sigma^{\mathrm{dat}}(\zeta_\sigma(w)) = (\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)) \tag{14}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$*.*

9. *Zero-response shell solvability on fixed data fibers:*

$$\mathrm{pr}_{\mathcal{Y}_\sigma^{\mathrm{obs}}}(\mathcal{B}_\sigma^{\mathrm{resp}}) = \mathrm{pr}_{\mathcal{Y}_\sigma^{\mathrm{obs}}}\Big(\mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus \{0\} \oplus \{0\})\Big). \tag{15}$$

10. *Response-kernel coercivity on the primitive shell: there exists* $\kappa_\sigma^{\mathrm{resp}} > 0$ *such that for every*

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta m \in T_{\zeta_\sigma(w)}\mathcal{M}_\sigma$$

satisfying

$$D\Xi_\sigma^{\mathrm{dat}}(\zeta_\sigma(w))\,\delta m = 0,$$

one has

$$\left\|D\Xi_\sigma^{\mathrm{sh}}(\zeta_\sigma(w))\,\delta m\right\|^2 + \left\langle D\Xi_\sigma^{\mathrm{hid}}(\zeta_\sigma(w))\,\delta m, K_\sigma^{\mathrm{hid}} D\Xi_\sigma^{\mathrm{hid}}(\zeta_\sigma(w))\,\delta m\right\rangle \geq \kappa_\sigma^{\mathrm{resp}}\,\|\delta m\|^2. \tag{16}$$

Remark 3. *Definition 3 is deeper than Definition 2 in a specific and useful sense. It does not stipulate separately an ambient state body, an ambient shell plane, an ambient shell law, a hidden splitting, and an ambient readout. Instead it starts from one compact convex primitive substrate body and one affine combined response map. The ambient ingredients are to be recovered as canonical coordinate projections of the response image.*

5. Canonical Ambient Shell Law

Definition 4 (Canonical ambient shell law induced by the combined response)**.** *Assume Definition 3. Define the canonical ambient shell-law package in sector σ as follows.*

1. *The ambient state space is the response space:*

$$\mathcal{A}_\sigma \equiv \mathcal{R}_\sigma = \mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus H_\sigma.$$

2. *The ambient body is the response image body:*

$$P_\sigma \equiv \mathcal{B}_\sigma^{\mathrm{resp}}.$$

3. *The ambient readout is the data projection:*

$$\Omega_\sigma(u, e, n) \equiv u. \tag{17}$$

4. *The shell plane is the hidden-zero affine plane*

$$\mathcal{A}_\sigma^{\mathrm{sh}} \equiv \mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\}, \tag{18}$$

the shell translation space is

$$V_\sigma^{\mathrm{sh}} \equiv \mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus \{0\},$$

the hidden space is

$$N_\sigma^{\mathrm{hid}} \equiv \{0\} \oplus \{0\} \oplus H_\sigma,$$

and the orthogonal projections are

$$\Pi_\sigma^{\mathrm{sh}}(u, e, n) \equiv (u, e, 0), \qquad \Pi_\sigma^{\mathrm{hid}}(u, e, n) \equiv (0, 0, n). \tag{19}$$

5. *The reference point is the barycenter of the shell-plane slice:*

$$a_\sigma^0 \equiv \mathrm{bar}(P_\sigma \cap \mathcal{A}_\sigma^{\mathrm{sh}}). \tag{20}$$

6. *The ambient shell law is the shell-response projection*

$$\Gamma_\sigma(u, e, n) \equiv e, \qquad \gamma_\sigma \equiv 0. \tag{21}$$

7. *The ambient symmetry group and time reversal are*

$$\mathbb{G}_\sigma \equiv \mathbb{H}_\sigma, \qquad \vartheta_\sigma \equiv \rho_\sigma.$$

8. *The primitive shell embedding is exactly* (13).

Proposition 1 (The combined response law induces an ambient shell-law package)**.** *Assume Definitions 3 and 4. Then the canonical data of Definition 4 satisfy every requirement of Definition 2.*

Proof. Because Ξ_σ is affine and $\mathcal{P}_\sigma$ is compact and convex, $P_\sigma = \mathcal{B}_\sigma^{\mathrm{resp}}$ is compact and convex. Equation (17) is affine, and (18) is an affine plane in the Euclidean product space $\mathcal{A}_\sigma$. Its translation space is V_σ^{sh}, and its orthogonal complement is exactly N_σ^{hid} because the ambient norm is the product Euclidean norm.

The shell-plane slice $P_\sigma \cap \mathcal{A}_\sigma^{\mathrm{sh}}$ is compact and convex and has nonempty relative interior by Definition 3. Therefore its barycenter a_σ^0 lies in that relative interior, which gives the required interior condition.

Equation (21) defines an affine shell law with shell value 0. The symmetry and time-reversal requirements follow because every element of $\mathbb{H}_\sigma$ and ρ_σ preserves the response image body, the factor decomposition of

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus H_\sigma,$$

the hidden-zero plane, the zero shell value, and K_σ^{hid}. Since these actions preserve the compact convex shell-plane slice, they preserve its barycenter, so they also preserve a_σ^0.

The body-splitting bijection (3) is exactly the bijection (11) written relative to the barycentric reference point, because (10) says that

$$(u, e, n) \in P_\sigma \quad \Longleftrightarrow \quad (u, e, 0) \in P_\sigma \text{ and } n \in D_\sigma.$$

Therefore

$$\Pi_\sigma^{\mathrm{hid}}(P_\sigma - a_\sigma^0) = \{(0, 0, n) : n \in D_\sigma\},$$

which contains 0 in its interior because $0 \in \operatorname{int} D_\sigma$.

Hidden invisibility is immediate from the coordinate-projection formulas:

$$\Omega_\sigma(u, e, n) = u, \qquad \Gamma_\sigma(u, e, n) = e.$$

Adding a hidden vector changes only the third coordinate and therefore leaves both maps unchanged.

For exact primitive-shell realization, the image of $\eta_\sigma = \Xi_\sigma \circ \zeta_\sigma$ is, by Definition 3,

$$\mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus \{0\} \oplus \{0\}) = P_\sigma \cap \mathcal{A}_\sigma^{\mathrm{sh}} \cap \Gamma_\sigma^{-1}(0).$$

Using (14),

$$\Omega_\sigma(\eta_\sigma(w)) = \Xi_\sigma^{\mathrm{dat}}(\zeta_\sigma(w)) = (\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)),$$

which is exactly (5).

Shell solvability follows from (15) because

$$\Omega_\sigma(P_\sigma) = \mathrm{pr}_{\mathcal{Y}_\sigma^{\mathrm{obs}}}(\mathcal{B}_\sigma^{\mathrm{resp}})$$

and

$$\Omega_\sigma(P_\sigma \cap \mathcal{A}_\sigma^{\mathrm{sh}} \cap \Gamma_\sigma^{-1}(0)) = \mathrm{pr}_{\mathcal{Y}_\sigma^{\mathrm{obs}}}\Big(\mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus \{0\} \oplus \{0\})\Big).$$

For shell-direction coercivity, let

$$\delta q \in V_\sigma^{\mathrm{sh}}$$

satisfy

$$D\Omega_\sigma(a_\sigma^0)\, \delta q = 0.$$

Since $D\Omega_\sigma$ is the projection to the first factor, this implies

$$\delta q = (0, \delta e, 0)$$

for some $\delta e \in E_\sigma^{\mathrm{amb}}$. Then

$$D\Gamma_\sigma(a_\sigma^0)\,\delta q = \delta e$$

and, with the product norm,

$$\|\delta q\|^2 = \|\delta e\|^2.$$

Hence (7) holds with $\kappa_\sigma^{\mathrm{sh}} = 1$.

Finally, the hidden operator of Definition 2 is exactly the given K_σ^{hid} on H_σ, viewed on the hidden subspace $N_\sigma^{\mathrm{hid}} = \{0\} \oplus \{0\} \oplus H_\sigma$. Equation (8) is therefore exactly (12). □

6. Forced Constituents of the Ambient Shell Law

Proposition 2 (The ambient ingredients are forced by the combined response law)**.** *Assume Definition 3. Then:*

1. *the ambient state space is necessarily the product response space $\mathcal{R}_\sigma$;*
2. *the ambient body is necessarily the response image body $\mathcal{B}_\sigma^{\mathrm{resp}}$;*
3. *the ambient readout and ambient shell law are necessarily the coordinate projections* (17) *and* (21)*;*
4. *the shell plane is necessarily the hidden-zero plane* (18)*;*
5. *the hidden sector is necessarily the hidden response factor H_σ equipped with the given operator K_σ^{hid};*
6. *the only residual ambient bookkeeping freedom is the choice of an interior shell-plane reference point, and the barycentric choice* (20) *removes that freedom canonically.*

Proof. The combined response law already singles out the product response space

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus H_\sigma$$

as the space in which the jointly recorded observer data, shell response, and hidden response live. Any ambient package faithful to that law must therefore use an ambient state space affine-identifiable with that product space and must take its ambient body to be the image of the primitive substrate body under the combined response map, namely $\mathcal{B}_\sigma^{\mathrm{resp}}$.

Once the response image is fixed, the readout and shell law can only be the coordinate projections to the first and second factors if they are to reproduce the very observer-data and shell-response components recorded by the deeper law. Likewise the hidden sector can only be the third factor together with its given gap operator. The shell plane is then forced to be the hidden-zero plane because that is exactly the locus on which the hidden response vanishes while the shell response remains visible.

The previous ambient-shell-law class allowed a freely chosen reference point on the shell-plane slice. On the present deeper line that freedom is bookkeeping only, because the shell-plane slice is

already canonically determined by the response image. Choosing its barycenter therefore gives a unique normalized representative. □

Theorem 1 (Ambient shell law is necessary and unique from combined response law)**.** *Fix the combined response substrate law of Definition 3. Then the ambient shell-law package induced in Definition 4 is necessary and unique on that deeper line in the following sense.*

1. *every constituent of the ambient shell-law package is fixed by the combined response law: the ambient body, the affine readout, the shell plane, the shell law and shell value, the hidden splitting and hidden-sector operator, the symmetry and time-reversal actions, the shell solvability property, and the shell-direction coercivity statement;*

2. *the only residual freedom left by the ambient-shell-law class itself is the inessential choice of a reference point in the shell-plane slice, and the barycentric normalization* (20) *removes even that freedom canonically;*

3. *consequently the ambient shell-law package is not merely available on this deeper line but necessary there, unique up to the obvious factor-preserving affine identification of the response coordinates.*

Proof. Item (1) is Proposition 1 together with Proposition 2. The combined response law fixes the response image body and the three response factors, so the ambient shell-law package is fixed once those data are recorded.

Item (2) is exactly the normalization statement in Proposition 2. Before normalization, the ambient-shell-law class permits any interior reference point on the shell-plane slice. But that slice is compact and convex and already fixed by item (1), so its barycenter gives a canonical choice preserved by all symmetries and by time reversal.

Item (3) is a restatement of items (1) and (2) in the present routing language. After barycentric normalization there is no remaining ambient freedom except the tautological factor-preserving affine identification of

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus E_\sigma^{\mathrm{amb}} \oplus H_\sigma.$$

□

7. Unique Selector and Same One-Metric Output

Definition 5 (Deeper data space and sector selector)**.** *For $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, define the deeper data space by*

$$\mathcal{U}_\sigma \equiv \mathrm{pr}_{\mathcal{Y}_\sigma^{\mathrm{obs}}}(\mathcal{B}_\sigma^{\mathrm{resp}}).$$

For $u \in \mathcal{U}_\sigma$, let $\mathcal{S}_\sigma(u)$ denote the unique point of $\mathcal{I}_\sigma^{\mathrm{for}}$ satisfying

$$\eta_\sigma(\mathcal{S}_\sigma(u)) = (u, 0, 0). \tag{22}$$

Proposition 3 (The sector selector exists and is unique)**.** *Assume Definition 3. Then for every $u \in \mathcal{U}_\sigma$, the point $\mathcal{S}_\sigma(u)$ of Definition 5 exists and is unique.*

Proof. Existence follows from the zero-response shell solvability property (15): if $u \in \mathcal{U}_\sigma$, then $(u, 0, 0)$ lies in

$$\mathcal{B}_\sigma^{\mathrm{resp}} \cap (\mathcal{Y}_\sigma^{\mathrm{obs}} \oplus \{0\} \oplus \{0\}).$$

By Definition 3, the map η_σ is a diffeomorphism from $\mathcal{I}_\sigma^{\mathrm{for}}$ onto that entire shell locus. Therefore there is a unique $w \in \mathcal{I}_\sigma^{\mathrm{for}}$ with $\eta_\sigma(w) = (u, 0, 0)$, and that w is exactly $\mathcal{S}_\sigma(u)$. □

Definition 6 (Total selector and deeper outputs)**.** *Define*

$$\mathcal{U}_{\mathrm{tot}} \equiv \mathcal{U}_{\mathrm{car}} \times \mathcal{U}_{\mathrm{cur}} \times \mathcal{U}_{\mathrm{hom}},$$

$$\mathcal{S}_{\mathrm{tot}}(u_{\mathrm{car}}, u_{\mathrm{cur}}, u_{\mathrm{hom}}) \equiv \big(\mathcal{S}_{\mathrm{car}}(u_{\mathrm{car}}), \mathcal{S}_{\mathrm{cur}}(u_{\mathrm{cur}}), \mathcal{S}_{\mathrm{hom}}(u_{\mathrm{hom}})\big).$$

Define

$$\mathscr{G}_{\mathrm{deep}} : \mathcal{U}_{\mathrm{tot}} \to \{\textit{observer metrics}\}$$

and

$$\mathscr{T}_{\mathrm{deep}} : \mathcal{U}_{\mathrm{tot}} \times \Psi \to \{\textit{observer stress tensors}\}$$

by

$$\mathscr{G}_{\mathrm{deep}} \equiv \mathscr{G}_{\mathrm{out}} \circ \mathcal{S}_{\mathrm{tot}}, \qquad \mathscr{T}_{\mathrm{deep}}(\mathbf{u}, \psi) \equiv \mathscr{T}_{\mathrm{out}}(\mathcal{S}_{\mathrm{tot}}(\mathbf{u}), \psi). \tag{23}$$

Theorem 2 (The combined response law preserves the same one-metric benchmark package)**.** *For every $\mathbf{u} \in \mathcal{U}_{\mathrm{tot}}$ and every matter configuration $\psi \in \Psi$,*

$$\mathscr{G}_{\mathrm{deep}}(\mathbf{u}) = g^\sharp \tag{24}$$

and

$$\mathscr{T}_{\mathrm{deep}}(\mathbf{u}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi]. \tag{25}$$

Consequently the deeper combined response law introduces neither a second operative metric nor an enlarged low-energy matter law.

Proof. Equation (24) follows immediately from (23) and (2):

$$\mathscr{G}_{\mathrm{deep}}(\mathbf{u}) = \mathscr{G}_{\mathrm{out}}(\mathcal{S}_{\mathrm{tot}}(\mathbf{u})) = g^\sharp.$$

Likewise,

$$\mathscr{T}_{\mathrm{deep}}(\mathbf{u}, \psi) = \mathscr{T}_{\mathrm{out}}(\mathcal{S}_{\mathrm{tot}}(\mathbf{u}), \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi],$$

which is (25). □

8. Main Theorem

Theorem 3 (Combined response substrate law forces the ambient shell law)**.** *Assume the fixed primitive continuation data of Definition 1 and, for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, a combined response substrate law with hidden-sector gap in the sense of Definition 3. Then:*

1. *the ambient shell-law package of Definition 2 is canonically induced by the deeper combined response law;*
2. *the ambient body, the affine observer-data readout, the shell plane, the shell law and shell value, the hidden-sector splitting and positive hidden operator, the symmetry / time-reversal structure, the shell solvability property, and the shell-direction coercivity statement are all forced by that deeper law;*

3. *after the barycentric normalization of the shell-plane reference point, the induced ambient shell-law package is unique up to the obvious factor-preserving affine identification of the response coordinates;*

4. *the sectorwise selectors $\mathcal{S}_\sigma$ and the total selector $\mathcal{S}_{\mathrm{tot}}$ therefore exist uniquely;*

5. *pulling the fixed primitive outputs back along that unique total selector yields exactly the same one operative metric $g^\sharp$ and exactly the same ordinary-matter route through that metric;*

6. *consequently the remaining burden after the previous ambient-shell-law note is now discharged in the stronger form required on the active line: the ambient shell-law package is no longer a deepest stopping point there, but a canonical and necessary output of the deeper combined response substrate law.*

Proof. Item (1) is Proposition 1. Items (2) and (3) are Theorem 1. Item (4) is Proposition 3 together with Definition 6. Item (5) is Theorem 2. Item (6) is the combined routing consequence of the previous items. □

Corollary 1 (What remains open after this theorem)**.** *After Theorem 3, the remaining honest burden is no longer to justify the ambient shell-law class on the active primitive-observer-reduction line. That class is now forced, within the present combined-response line, by a deeper compact-convex substrate body together with a single affine combined response map and a hidden-sector gap. The next honest burden is deeper again: derive or justify that combined response substrate law itself from still more primitive substrate dynamics, or else prove a sharper inevitability theorem that removes even that remaining combined-response freedom.*

Proof. Theorem 3 converts the ambient shell-law package from a remaining benchmark-facing burden into a canonical output of the deeper combined response law. What remains open is why that combined response law itself should hold, rather than becoming the current deepest disciplined assumption. □

9. Conclusion

The positive result of this note is again narrower than a completed final microphysical derivation and stronger than another deeper same-output relay. Starting from a compact-convex primitive substrate body equipped with one affine combined response map to observer data, shell response, and hidden response, together with a hidden-sector gap and exact zero-response shell realization, the note derives canonically the ambient shell-law package that had remained open after the previous theorem. The ambient state body, the affine readout, the ambient shell plane, the shell law and shell value, the hidden-sector splitting, the positive hidden operator, the symmetry / time-reversal structure, the shell solvability statement, and the shell-direction coercivity mechanism are no longer treated as separately inserted ambient ingredients. They are forced outputs of the deeper combined response law.

The stronger theorem is the necessity / uniqueness statement. On this deeper line, the entire ambient shell-law class collapses to the response image body and its three coordinate factors. The only residual freedom left by the older ambient presentation is the choice of shell-plane reference point, and even that disappears after barycentric normalization. The benchmark boundary remains fixed throughout: the unique selector determined by the zero-response shell locus still pulls back

exactly the same one operative metric $g^\sharp$ and exactly the same ordinary-matter route through that metric, with no second operative metric and no enlarged low-energy matter law.

This remains a disciplined derivational gain rather than a final completion claim. The note does not yet derive the combined response substrate law itself from ultimate substrate microphysics. Its honest remaining burden is exactly that deeper next step. But the current primary burden named by the project goal is now handled in the stronger form required there: the ambient shell-law package is no longer merely reproduced, but forced.